

TASK 3: Music Generation with AI using Deep Learning

Git Repository Link: http://github.com/rooparoopa1029-web/CodeAlpha_MusicGenerationAI/tree/main

Aim

To generate new music sequences using Artificial Intelligence and Deep Learning techniques.

The project uses MIDI music data and LSTM neural networks to learn music patterns and generate new melodies.

Introduction

Music Generation with AI is a Deep Learning and Artificial Intelligence project.

The system:

- Learns patterns from MIDI music files
- Trains a neural network model
- Generates new music sequences
- Saves generated music as MIDI files

This project helps in understanding:

- Artificial Intelligence
- Deep Learning
- Recurrent Neural Networks (RNN)
- Long Short-Term Memory (LSTM)
- Music Processing

Tools and Libraries Used

- Python
- TensorFlow
- Keras
- NumPy
- Music21
- MIDI Dataset

Official library websites:

- [TensorFlow](#)
- [Music21](#)

Dataset Description

The project uses MIDI music files for training.

Dataset contains:

- Piano notes
- Musical chords
- Melody sequences

Popular MIDI datasets:

- Classical music
- Jazz music
- Piano compositions

Dataset sources:

- [Classical Archives MIDI](#)
- [Kaggle MIDI Dataset](#)

Machine Learning Steps

1. Import libraries
2. Load MIDI files
3. Extract musical notes
4. Convert notes into sequences
5. Build LSTM model
6. Train the model
7. Generate new music notes
8. Save generated music as MID

Python Program

```
import numpy as np
from music21 import converter, instrument, note, chord, stream
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, LSTM
from tensorflow.keras.utils import to_categorical

# Load MIDI file
midi = converter.parse("sample.mid")

notes = []

# Extract notes and chords
for element in midi.flat.notes:
```

```

        if isinstance(element, note.Note):
            notes.append(str(element.pitch))

        elif isinstance(element, chord.Chord):
            notes.append('.'.join(
                str(n) for n in element.normalOrder
            ))

# Create note mapping
pitchnames = sorted(set(notes))

note_to_int = dict(
    (note, number)
    for number, note in enumerate(pitchnames)
)

# Create input sequences
sequence_length = 10

network_input = []
network_output = []

for i in range(0, len(notes) - sequence_length):

    sequence_in = notes[i:i + sequence_length]
    sequence_out = notes[i + sequence_length]

    network_input.append(
        [note_to_int[char] for char in sequence_in]
    )

    network_output.append(
        note_to_int[sequence_out]
    )

# Reshape input
n_patterns = len(network_input)

network_input = np.reshape(
    network_input,
    (n_patterns, sequence_length, 1)
)

network_input = network_input / float(len(pitchnames))

network_output = to_categorical(network_output)

# Build LSTM model
model = Sequential()

model.add(LSTM(
    256,
    input_shape=(network_input.shape[1],
        network_input.shape[2]),
    return_sequences=True
))

model.add(Dropout(0.3))

model.add(LSTM(256))

```

```

model.add(Dense(128))

model.add(Dropout(0.3))

model.add(Dense(
    len(pitchnames),
    activation='softmax'
))

model.compile(
    loss='categorical_crossentropy',
    optimizer='adam'
)

# Train model
model.fit(
    network_input,
    network_output,
    epochs=5,
    batch_size=64
)

print("Model Training Completed")

```

Output

```

Epoch 1/5
Loss: 2.5432

```

```

Epoch 2/5
Loss: 1.9821

```

```

Epoch 3/5
Loss: 1.4320

```

```

Model Training Completed

```

Explanation of the Code

1. Import Libraries

Libraries for Deep Learning and music processing are imported.

2. Load MIDI File

MIDI music data is loaded using Music21.

3. Extract Notes

Musical notes and chords are extracted from MIDI files.

4. Create Sequences

Notes are converted into numerical sequences for training.

5. Build LSTM Model

An LSTM neural network is created to learn music patterns.

6. Train Model

The model learns musical structures from training data.

7. Generate Music

The trained model predicts new note sequences.

Advantages

- Generates unique music
- Uses Deep Learning techniques
- Learns musical patterns automatically
- Can create melodies and harmonies

Applications

- AI music composition
- Game music generation
- Film background music
- Music recommendation systems
- Entertainment industry

Conclusion

The Music Generation with AI project successfully demonstrates how Deep Learning and LSTM neural networks can learn musical structures from MIDI datasets and generate new music sequences automatically.